

Collapsing Decision Boundaries

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Introduction

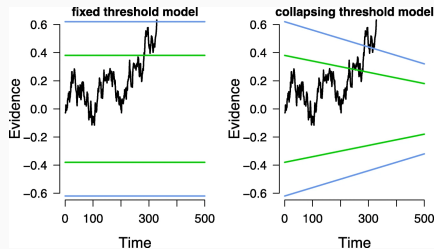
- **People make fast decisions every day:**
 - “Was that a green or red light?”
 - “Do I press left or right?”
- **In psychology experiments:**
 - People are asked to make quick choices.
 - We record **what they choose** and **how long it takes**.
- **Speed-Accuracy Trade-off model:**
 - How do people change how they decide when they're under time pressure?
- **We use models to simulate this decision process.**
 - The model acts like a brain: it builds up “evidence” and then makes a choice.

Background & Motivation

Key Findings – Katsimpokis et al. (2020):

- **Fixed decision boundaries** fit best when participants were simply *instructed to respond quickly*.
- **Linearly collapsing boundaries** provided a better fit when an *explicit deadline* was imposed.

Figure 1: Different assumptions in diffusion models



What remains unexplored?

- **Non-linear boundary collapse:**
 - e.g., *exponential, sigmoid*, or other curved functions.
- **Urgency gating mechanisms:**
 - Rising urgency over time that *modulates the evidence accumulation process*.

Research Question:

- Can *non-linear boundaries* or *urgency gating* improve model fit and explain behavior under time pressure more accurately?

The Evidence Accumulation Model – Drift Diffusion Model (DDM)

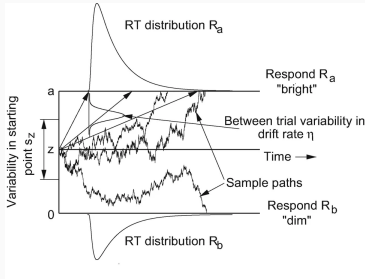


Figure 2: Drift Diffusion Model

- **Drift Diffusion Model (DDM):**

- Evidence accumulation equation:

$$dx = \nu \cdot \Delta t + \sqrt{\Delta t} \cdot \mathcal{N}(0, 1)$$

- A decision is made once the evidence hits either the upper or lower boundary.

- **Key DDM Parameters:**

- ν – Drift rate (evidence quality)
- a – Boundary separation (decision threshold)
- a_0 – Starting point (bias)
- We have considered a very simple DDM without taking into account inter-trial drift variability or variability of the starting point.

Exponential Boundary Collapse

- **Why use non-linear collapse?**
 - Linear boundaries assume constant pressure to decide.
 - Human decision pressure likely grows *non-linearly* under deadlines.
 - Exponential collapse allows more flexibility: slower early decline, sharper drop later.
- **Exponential boundary function:**

$$a(t) = a_0 \cdot e^{t/\tau}$$

- a_0 : Initial boundary height
- τ : Decay rate
- t : Decision time

BayesFlow Model Specifications

- The inference parameters: ν , $\alpha(a_0)$, τ
- Priors: $\nu \sim \text{Gamma}(k = 10, \theta = 0.25)$, $\alpha \sim \text{Gamma}(k = 10, \theta = 0.05)$,
 $\tau \sim \text{Uniform}(-3, -0.5)$
- Inference conditions (summary variables): Mean Standard Deviation, Quantiles (20%, 50%, 80%), p_{response}
- Workflow: Inference network - Flowmatching (default conditions), no summary network
- Training condition - Offline fit, 30-60 epochs, batch size: 500

DDM Inference Diagnostics – Parameter Estimation Quality

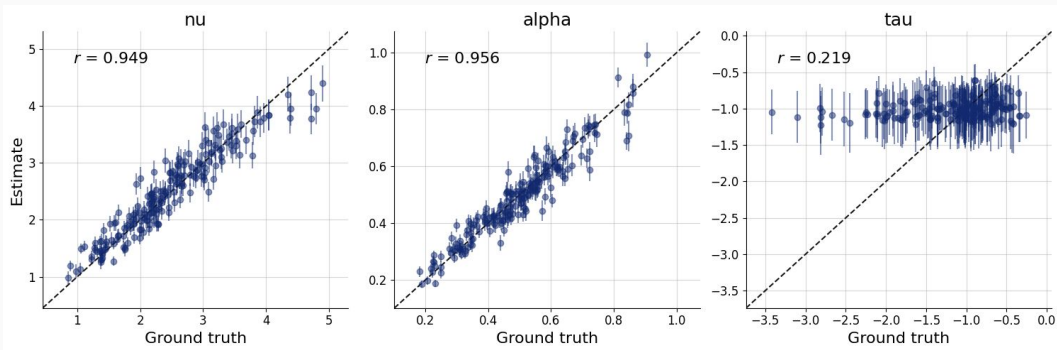


Figure 3: Parameter Recovery

DDM Inference Diagnostics – Parameter Estimation Quality - Contd

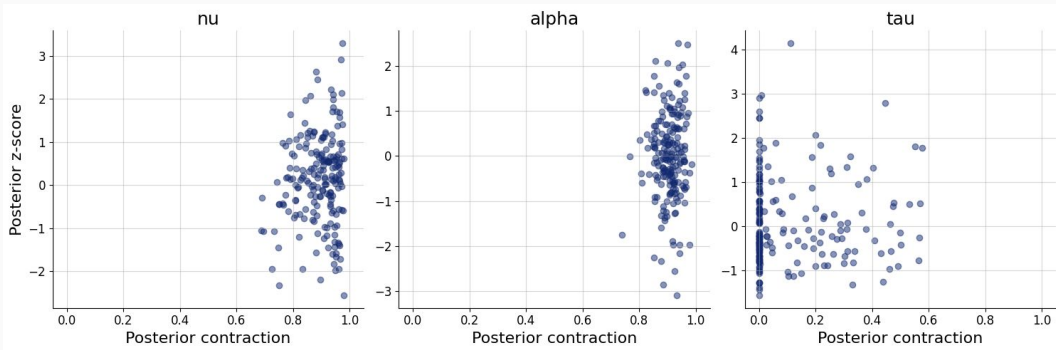


Figure 4: Posterior Contraction vs. z-scores

DDM Inference Diagnostics – Parameter Estimation Quality - Contd

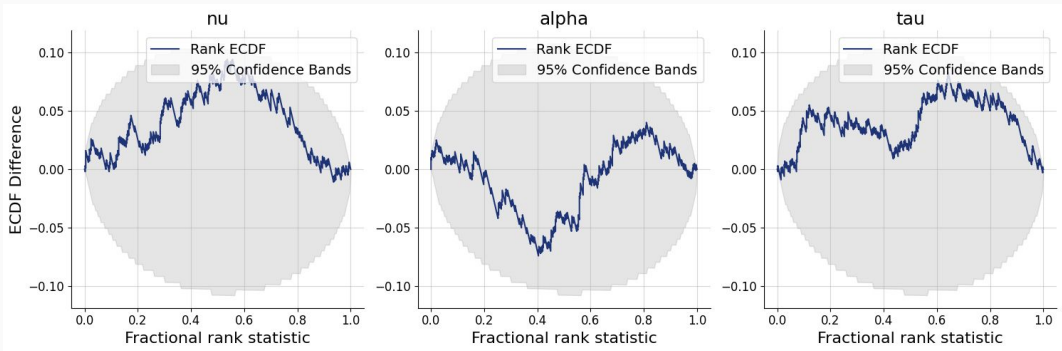


Figure 5: SBC-ECDF Difference

DDM Inference Diagnostics – Parameter Estimation Quality - Contd

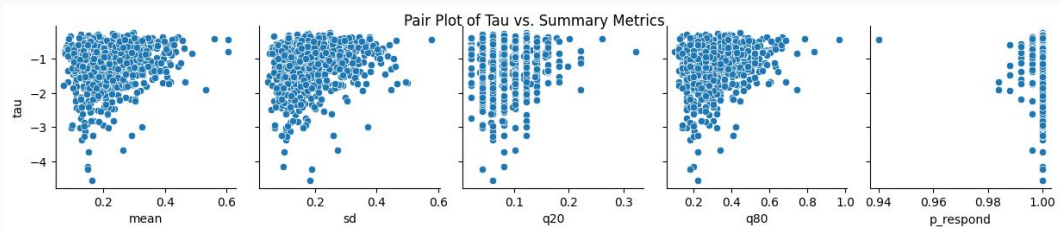


Figure 6: Tau vs Summary metrics

- We implemented a **simple DDM with an exponentially collapsing threshold** to explain the speed-accuracy trade-off in two-choice tasks.
- **Parameter recovery results:**
 - **Drift rate (ν)** and **boundary height (α)** were reasonably recovered.
 - **Decay rate (τ)** was poorly identified.
- **Why is τ hard to recover?**
 - Possible *correlation between α and τ* : different parameter combinations can produce similar behavior.
 - The current summary statistics may not contain enough information to disambiguate τ .
- **Next Steps:**
 - Explore better summary features or switch to raw RT distributions.
 - Test the model on real experimental data.

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